

Analyzing Firm-Level Innovation and Spillover Effects Using Econometric Methods

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Abstract

The relationship between innovative investment and innovation output has always been a concern for entrepreneurs and economists. This paper examines the relationship between R&D investment, technology spillovers, and patent output at the firm level using an eight-year panel dataset from 181 firms. To address potential biases due to unobserved heterogeneity and endogeneity, four econometric models are estimated and compared: Pooled OLS, fixed effects, random effects, and two-stage least squares instrumental variables (2SLS-IV). The results show that both R&D expenditures and technology spillovers have significant positive effects on patenting behavior, but the magnitude of these effects varies across models. The Hausman test indicates that the fixed effects model is superior to the random effects model, and using IV estimation to account for endogeneity yields different parameter estimates. The results emphasize the importance of model selection in assessing innovation drivers and demonstrate that both internal and external knowledge sources are crucial for firm-level innovation outcomes.

Introduction

Over the long term, innovation has been one of the key driving forces of economic development. Many companies and organizations around the world have achieved tremendous economic profits through unprecedented innovations, thereby stimulating the economic growth of related industries. Understanding the determinants of innovation is at the core of the economics of technological change. One of the most extensively studied indicators of innovation output has been the number of patent applications filed by firms. Among the main drivers of patent activity, research and development (R&D) expenditures have received particular attention due to their role in advancing new technologies. However, despite the intuitive relationship between R&D intensity and patent output, establishing causality empirically remains challenging due to issues such as simultaneity, unobserved heterogeneity, and the nonlinear nature of patent count data. In addition, another important factor affecting the innovative performance of enterprises is technology spillovers, the positive external influence from the R&D activities of other enterprises. Such spillover effects may come from observing competitors' innovation, labor mobility, reverse engineering, or shared infrastructure. Recognizing the role of spillovers is crucial because firms do not innovate in isolation; They rely more on mutual communication and learning, so that their knowledge production functions can be embedded in a wider technology network.

To examine these questions, this paper revisits the relationship between R&D, technology spillovers, and patents using a unique firm-level panel dataset constructed by Cincera (1997), covering 181 international manufacturing firms from 1983 to 1991. The initial study applied Poisson and GMM counting panel models to estimate the determinants of patent applications. On this basis, this study proposes a different econometric strategy that considers both firm-specific fixed effects and the potential endogeneity of R&D. Specifically, we adopt the fixed effects method to remove the invariant individual characteristics at the firm level and further adopt 2SLS-IV model and to deal with the endogeneity of R&D investment.

This paper is organized as the following: Section 2 describes the dataset details, including how key variables such as patents, R&D, and technological spillovers are constructed, and outlines the sample's key descriptive features; Section 3 presents the econometric models and identification strategies; Section 4 reports the empirical findings, including baseline estimates and robustness checks across different model specifications; Section 5 concludes by summarizing the main contributions of the paper and discussing the implications for innovation policy. By improving on the econometric specification and incorporating techniques to isolate causal relationships, this study will contribute to a deeper understanding of how internal and external knowledge sources shape firm-level innovation.

Description of the Data

This study uses a panel dataset first constructed by Cincera (1997) to examine the determinants of patenting behavior among large international manufacturing firms. The sample includes 181 companies from four major geographic regions (the United States, the European Union, Japan, and the rest of the world), spanning the period from 1983 to 1991. These firms were selected based on their continuous involvement in research and development (R&D) activities, with the inclusion criterion being the availability of data for at least nine consecutive years. This dataset is particularly well-suited for studying the innovation process, as it combines firm-level data such as annual R&D expenditures, patent applications, and industry affiliations with external source measurements of technology spillovers and geographic/technological opportunity structures.

The dependent variable in the analysis is the number of patent applications filed by each firm annually. These counts were obtained from the European Patent Office (EPO) database and reflect only patent filings, not grants, to better capture innovation intent. The primary explanatory variable is R&D investment, deflated by each country's gross domestic product (GDP) and converted into constant 1990 U.S. dollars to ensure comparability across firms and countries.

To more comprehensively evaluate the spillover effects of agency technology, we adopt

the sectoral analysis method proposed by Mohnen (1991). This method analyzes by calculating the proportion of R&D investment within specific sectors, using data sourced from the OECD's Anberd database, which covers the R&D activities of all companies within the sector and the company's own R&D investment. Through this intra-industry indicator, we can assume that companies within the same industry can obtain symmetrical benefits from each other's R&D efforts. This assumption is based on the common phenomenon of knowledge sharing and technology diffusion within the industry. While intra-industry indicators provide a concise analytical framework, academia has also developed more sophisticated measures of technology spillovers. For example, some studies quantify technology spillovers based on the proximity of patent category spaces, a method that can more accurately capture the path and intensity of technology transfer. Therefore, in the current study, we have chosen the intra-industry indicator, a relatively mature and easily comparable method, to balance the depth and breadth of research and ensure the feasibility and consistency of analysis. Industry classification is based on the Standard Industrial Classification (SIC) scheme, covering fifteen distinct manufacturing sectors.

Table 1 reports the descriptive statistics for key variables used in the analysis. The number of patents increases steadily over time, with the mean number of patents rising from 44 in 1983 to about 80 in 1990, before slightly decreasing in 1991. R&D intensity, as measured by the log of R&D expenditure, also shows a general upward trend, suggesting an increase in innovation-related investment across firms during the sample period.

Table 1
Descriptive Statistics

Statistic	Mean	St. Dev.	Min	Median	Max
firm_id	91.00	52.39	1	91	181
sector	6.32	4.24	1	5	15
region	2.64	0.74	1	3	4
patents_1983	43.99	91.76	0	11	596
patents_1984	48.70	97.89	0	14	693
patents_1985	50.61	101.63	0	17	789
patents_1986	55.55	110.32	0	19	897
patents_1987	57.36	111.17	0	17	871
patents_1988	64.22	119.79	0	19	891
patents_1989	73.31	136.40	0	21	839
patents_1990	79.75	154.20	0	24	922
patents_1991	73.59	150.95	0	20	925
log_RD_1983	4.93	1.27	0.87	4.84	7.94
log_RD_1984	5.04	1.24	2.73	4.91	8.20
log_RD_1985	5.13	1.26	3.05	5.01	8.50
log_RD_1986	5.18	1.23	3.18	5.00	8.31

log_RD_1987	5.23	1.24	2.95	5.10	8.33
log_RD_1988	5.29	1.25	3.27	5.18	8.37
log_RD_1989	5.31	1.28	3.08	5.17	8.67
log_RD_1990	5.35	1.28	3.15	5.18	8.70
log_RD_1991	5.35	1.30	3.14	5.13	8.47
log_spillover_1983	9.19	0.88	6.87	9.23	10.42
log_spillover_1984	9.27	0.89	6.90	9.43	10.48
log_spillover_1985	9.35	0.91	6.88	9.57	10.57
log_spillover_1986	9.37	0.92	6.82	9.56	10.59
log_spillover_1987	9.40	0.94	6.91	9.56	10.63
log_spillover_1988	9.45	0.94	7.02	9.69	10.66
log_spillover_1989	9.48	0.93	7.08	9.78	10.66
log_spillover_1990	9.53	0.94	7.09	9.83	10.71
log_spillover_1991	9.56	0.92	7.11	9.85	10.76

Note. Descriptive statistics for all variables are based on firm-year observations. Patent counts refer to the number of patents granted in each year from 1983 to 1991. log_RD denotes the natural logarithm of R&D expenditures for each corresponding year. Values are reported as means, standard deviations, minimums, medians, and maximums. The data show increasing trends in both patenting activity and R&D investment over time, with substantial variation across firms.

Description of the Models

Pooled OLS model

The first model we used is the Pooled Ordinary Least Squares (Pooled OLS) model which treats all observations in panel data as independent and identically distributed cross-sectional data points, ignoring structural differences between firms and time. Its basic form is as follows:

$$Y_{it} = \beta_0 + \beta_1 \log(RD_{it}) + \beta_2 \log(Spillover_{it}) + u_{it} \quad (1)$$

where Y_{it} represents the number of patents that i th enterprise applied at year t ; $\log(RD_{it})$ represents the enterprise's R&D investment in that year (taking the logarithm); $\log(Spillover_{it})$ represents the intensity of technological spillover in the industry or region where the enterprise is located (also taking the logarithm); c_i captures unobserved, time-invariant firm heterogeneity, such as baseline productivity or management quality, and u_{it} is an idiosyncratic error; β_0 is the overall intercept, which represents the average value of patents at all enterprises and all time points when the explanatory variable takes the value of 0; β_1 is the coefficient that reflects the marginal output effect of the enterprise's R&D investment, that is, for every 1 unit (logarithmic) increase in R&D investment, the expected increase in patent output would be β_1 units; β_2 reflect the knowledge spillover effect from other enterprises or industries.

Pooled OLS is the most fundamental method for panel data analysis and holds

significant reference value in the initial stage of model construction. It provides preliminary estimates of the relationships between variables, serving as a comparative basis for more complex models such as fixed effects and random effects. Its estimation is simple and easy to implement, without the need for individual or time-level transformations or controls. It can preliminarily assess whether R&D investment and technology spillovers have statistical impacts on patent output.

This model assumes that there are no systematic individual or temporal effects in the data, and it also must satisfy a series of key assumptions:

- (1) Linearity: The relationship between the dependent variable and the independent variables is linear.
- (2) There is no linear or nonlinear correlation between the explanatory variable and the error term. This can be written as:

$$E[u_{it}|X_{it}] = 0$$

- (3) Homoskedasticity: The error terms of all observations have the same variance, that is:

$$Var(u_{it}) = \sigma_u^2 \quad \text{for any } i, t$$

- (4) There is no correlation between the error terms of the same company in different years. This can be written as:

$$Cov(u_{it}, u_{is}) = 0 \quad \text{for any } t \neq s$$

- (5) There is no complete multicollinearity among the explanatory variables, which forces the explanatory variable matrix to have full rank.
- (6) The error term has no autocorrelation and no heteroscedasticity. This can be written as:

$$u_{it} \sim i.i.d(0, \sigma_u^2)$$

In summary, pooled OLS provides a fast and intuitive way to explore the correlation between variables, serving as a starting point or benchmark model in panel data analysis. However, due to its failure to control individual and time heterogeneity, the results may be biased. Therefore, in this study, we further use models such as Fixed Effects, Random Effects, and 2SLS for robust testing and comparison.

Random Effects Model

To examine the impact of research and development investment ($\log(RD)$) and technology spillover ($\log(Spillover)$) on patent output ($patents$), while controlling for heterogeneity to some extent, I estimated a random effects (RE) panel model. This model will combine cross-sectional and time-series information, and is suitable for improving estimation efficiency when the explanatory variables are not perfectly correlated with individual effects. The basic form of the random effects model is:

$$Y_{it} = c_i + \beta_0 + \beta_1 \log(RD_{it}) + \beta_2 \log(Spillover_{it}) + u_{it} \quad (2)$$

Why did I choose the random effects model? Because it can estimate the effects of time-invariant variables. In this study, the heterogeneity between companies may not be completely strongly correlated with $\log(RD)$ and $\log(Spillover)$, so RE is an appropriate model for exploring whether we can balance heterogeneity control and

efficiency improvement.

The main assumptions of the Random Effects model include:

- (1) Explanatory variables are independent of random effects. This means that unobserved traits at the company level (such as an innovative culture) do not systematically affect the explanatory variables. It can be written as:

$$Cov(X_{it}, c_i) = 0$$

- (2) The random effect is independent of the error term, that is:

$$Cov(c_i, u_{it}) = 0$$

- (3) The error term has no autocorrelation and no heteroscedasticity. This can be written as:

$$u_{it} \sim i.i.d(0, \sigma_u^2)$$

The significance of these assumptions is that they ensure the consistency and efficiency of model estimation. When there is correlation between explanatory variables and individual effects, RE estimation will be biased. The results will be shown in the next section.

Fixed Effects Model with Within Estimator

To better analyze the relationship between input-output variables at the enterprise level, with particular attention to the changing trends of the enterprise itself in different periods, the fixed effect model of within is used. The basic form of this model is:

$$Y_{it} = c_i + \beta_1 \log(RD_{it}) + \beta_2 \log(Spillover_{it}) + u_{it} \quad (3)$$

For each enterprise i th take the average in the time dimension and then subtract these two formulas to obtain the model after "mean removal", which is our fixed effects model with within estimator:

$$\begin{aligned} Y_{it} - \bar{Y}_i &= (c_i - \bar{c}_i) + \beta_1 (\log(RD_{it}) - \overline{\log(RD_i)}) \\ &\quad + \beta_2 (\log(Spillover_{it}) - \overline{\log(Spillover_i)}) \\ &\quad + (u_{it} - \bar{u}_i) \end{aligned} \quad (4)$$

It can be written as:

$$\check{Y}_{it} = \beta_1 \log(\check{RD}_{it}) + \beta_2 \log(\check{Spillover}_{it}) + \check{u}_{it} \quad (5)$$

where $\check{Y}_{it} = Y_{it} - \bar{Y}_i$ (and similarly for other regressors). In the fixed-effect model presented with formula (5), the significance of the β coefficient is similar to (3), but the level of interpretation has changed. It no longer describes cross-sectional differences among firms, but rather within-individual variations within the same firm at different times. The number of patents of the enterprise in that year is expected to be

approximately β_1 (β_2 respectively) more than its average patent output when its $\log(RD_{it})$ ($\log(Spillover_{it})$ respectively) of a certain year is one unit higher than its average. The fixed effect model eliminates traits such as enterprise scale, industry, and management ability that remain unchanged over time but may affect the dependent variable. Therefore, β reflects a dynamically changing causal relationship rather than a static level of comparison.

By using Hausman test, we are able to determine whether within model or random model is consistent and effective. The Hausman test results shown the within model is better. It implies that in the enterprise panel data, individual fixed characteristics (such as scale, management, industry characteristics, etc.) and explanatory variables are correlated, it is necessary to control these unchanging factors within the enterprises themselves. The test results will be shown in the next section.

Compared to random effects models, fixed effects models can effectively control unobservable time-invariant characteristics. When firm-specific traits are correlated with independent variables, using random effects model for estimation will result in biased estimators. By subtracting means, fixed effects models can effectively eliminate the impact of firm-specific traits on independent variables, thereby reducing endogeneity bias and obtaining more accurate estimation results. Another significant advantage of this model is its ability to fully utilize the structural advantages of panel data. By observing variable changes within individuals, fixed effects models can better capture the dynamic impact relationship between independent variables and dependent variables.

To ensure the validity and consistency of the estimators, the fixed effects model relies on the following key assumptions:

- (1) Linearity: It assumes a linear relationship between the dependent variable and the explanatory variables, thus there are no terms involving higher powers or cross products in the model.
- (2) Strict Exogeneity: It assumes that the error term should be conditionally independent of all time periods' explanatory variables at any given time. This assumption implies that firms' future or past decisions are not affected by current random shocks. It can be written as:

$$E(u_{it} | \log(RD_{i1}), \dots, \log(RD_{iT}), \log(Spillover_{i1}), \dots, \log(Spillover_{iT}), c_i) = 0$$

- (3) No Serial Correlation: It assumes that the error terms of firms are uncorrelated across different years, avoiding estimation bias caused by dynamic dependence. It can be written as:

$$E(u_{it}u_{is}) = 0 \text{ for } t \neq s$$

- (4) Homoskedasticity: It assumes that within individuals, the error terms have constant variance. If heteroscedasticity exists, robust adjustments to standard errors are needed. It can be written as:

$$Var(u_{it} | \log(RD_{it}), \log(Spillover_{it}), c_i) = \sigma^2$$

- (5) This model allows individual effects c_i to be correlated with explanatory variables.

This is the core advantage of fixed-effects models compared to random-effects models. This assumption holds that firm-specific time-invariant characteristics (such as governance level and technological foundation) may be correlated with explanatory variables like R&D investment, and thus cannot be treated as random errors. By removing firm-specific time-invariant characteristics, fixed-effects models avoid omitted variable bias. It can be written as:

$$\begin{aligned} Cov(c_i, \log(RD_{it})) &\neq 0 \\ Cov(c_i, \log(Spillover_{it})) &\neq 0 \end{aligned}$$

2SLS Model with Instrumental Variable

In the previous section, we used the fixed Effects (within) model to estimate the impact of enterprise R&D investment and knowledge spillover on patent output, controlling for the fixed and unobserved heterogeneity of enterprises. However, the fixed-effect model may still have an important problem: endogeneity bias. For instance, if certain unobserved time-varying factors (such as technological changes and policy shocks) simultaneously affect R&D investment and patent output, it will lead to the independent variable ($\log_RD$) is related to the error term and undermines the consistency of the estimation. To solve this problem, we introduced the instrumental variable method (2SLS). I have constructed the following two-stage least squares (2SLS) model, which is nested within the fixed effects framework:

First stage:

$$\begin{aligned} \log(\widehat{RD}_{it}) &= \gamma_1 \log(RD_{i,t-1}) + \gamma_2 \log(Spillover_{i,t-1}) + \gamma_3 \log(Spillover_{it}) + c_i \\ &+ \epsilon_{it} \end{aligned} \quad (4)$$

$$\begin{aligned} \log(\widehat{Spillover}_{it}) &= \delta_1 \log(RD_{i,t-1}) + \delta_2 \log(Spillover_{i,t-1}) + \delta_3 \log(Spillover_{it}) + c_i \\ &+ \epsilon_{it} \end{aligned} \quad (5)$$

Second stage:

$$Y_{it} = \beta_1 \log(\widehat{RD}_{it}) + \beta_2 \log(\widehat{Spillover}_{it}) + c_i + u_{it} \quad (6)$$

where $\log(\widehat{RD}_{it})$ and $\log(\widehat{Spillover}_{it})$ are the purified variable predicted by instrumental variables. $\log(RD_{i,t-1})$ and $\log(Spillover_{i,t-1})$ are the $\log(RD)$ and $\log(spillover)$ from last year, as an instrumental variable, they satisfies the correlation but does not directly affect the current error term. $\log(Spillover_{it})$ is

special since it appears in both functions in the first stage. It is one of the instrumental variables and also enters the structural equation as an explanatory variable. It is used as an already exogenous variable: an explanatory variable included in the set of instrumental variables. Since spillover is assumed to be not related to the error term and it can effectively explain the dependent variable. Therefore, it can appear as an explanatory variable in the structural equation. Because it is not regarded as an endogenous variable causing the problem by itself, its "dual role" does not violate the assumptions of the model.

Endogeneity issue is a critical point to consider when selecting models. This might lead to a potential correlation between the error term in our model and $\log(RD)$, thereby affecting our estimation results. To address this issue, the instrumental variable method provides an effective approach. This method introduces exogenous variables, such as lag variables, that are correlated with the explanatory variables but not with the error term, to conduct two-stage least squares (2SLS) estimation. Through this approach, the instrumental variable method can obtain consistent estimation results, effectively correcting biases caused by endogeneity. This makes our model more accurate and reliable, better reflecting the true relationships between variables.

To ensure the validity and consistency of the estimators, the 2SLS IV model relies on the following key assumptions:

- (1) Instrument variables such as $\log(RD_{i,t-1})$ must be significantly correlated to the endogenous explanatory variable $\log(RD_{it})$. This can be written as:

$$Cov(Z_{it}, X_{it}) \neq 0$$

where Z_{it} represents the instrument variables and X_{it} represents the explanatory variable.

- (2) Instrumental variables cannot be related to the structural error term. This can be written as:

$$Cov(Z_{it}, u_{it}) = 0$$

- (3) The error term is not correlated with the instrumental variables, explanatory variables and individual fixed effects in all periods. This can be written as:

$$E[u_{it}|c_i, Z_{is}, X_{is}] = 0 \text{ for all } s, t$$

- (4) There is no complete linear correlation between the explanatory variables or instrumental variables.

- (5) $E[u_{it}] = 0$.

So far, I have employed four models (Pooled OLS, Fixed Effects, Random Effects, 2SLS/IV) for estimation and comparison, which can enhance the study's rigor, the credibility of conclusions, and the reliability of causal inference. Below, I will present the results of these four models and conduct comparative analysis and explanation.

Presentation of Fitted Models and Interpretation

To examine the relationship between R&D investment, technology spillovers, and innovation output (measured by the number of patents) at the firm level, we estimated and compared four econometric models: pooled OLS, fixed effects (FE), random effects (RE), and two-stage least squares instrumental variables (2SLS-IV). Table 2 presents the results of these models.

Table 2

Comparison of Panel Data Models

Dependent variable:				
	Patents			
	Pooled OLS	Fixed Effects	Random Effects	2SLS-IV
	(1)	(2)	(3)	(4)
log_RD	55.519*** (2.326)	16.322*** (4.161)	31.326*** (3.411)	21.181*** (7.245)
log_spillover	-4.290 (3.158)	65.043*** (8.859)	27.705*** (6.096)	74.766*** (12.074)
Constant	-187.334*** (28.021)		-362.546*** (52.405)	
Observations	1,448	1,629	1,629	1,448
R ²	0.303	0.086	0.098	0.082
Adjusted R ²	0.302	-0.029	0.097	-0.050
F Statistic	313.434*** (df = 2; 1445)	68.022*** (df = 2; 1446)	176.856***	110.971***

Note:

*p<0.1; **p<0.05; ***p<0.01

Note. This table compares four econometric specifications estimating the effect of R&D investment (log_RD) and technological spillovers (log_spillover) on firm patenting activity. The models include pooled OLS, fixed effects, random effects, and two-stage least squares (2SLS) instrumental variable regression.

Among all models, the logarithm of R&D expenditures shows a significant positive correlation with patent output. However, the magnitude of the coefficients varies greatly. The pooled OLS estimated coefficient is very large (55.52), possibly reflecting an upward bias due to unobserved firm heterogeneity. The FE model significantly reduces this effect (16.32), suggesting that time-invariant firm characteristics, such as R&D culture or baseline innovation, can explain part of the observed relationship in the pooled model. The RE and 2SLS-IV models yield intermediate values (31.33 and 21.18, respectively), with 2SLS designed to address potential endogeneity issues in R&D

investment. The coefficient of the logarithm of spillovers (technological spillovers) also shows significant differences across models, with a small and negative coefficient (-4.29) in pooled OLS, which is statistically insignificant. In contrast, all other models show strong, positive, and significant effects, with the 2SLS-IV model estimating the largest effect (74.77). This suggests that spillover effects may be confounded by unobserved firm characteristics in pooled OLS, and become more apparent once heterogeneity and potential endogeneity are addressed. Although the Pooled OLS model reported the highest R^2 (0.303), this value may be overstated due to ignoring variable bias and failing to consider panel structure. The Fixed Effects and 2SLS-iv models reported lower R^2 values (0.086 and 0.082), but they provide more credible estimates by controlling for unobserved heterogeneity (FE) or addressing endogeneity (2SLS). The Random Effects model lies in between in model fit ($R^2 = 0.098$), although its applicability depends on the validity of the assumption that individual effects are uncorrelated with regressors. Importantly, the Fixed Effects model isolates variation within firms, which is particularly valuable for understanding changes in R&D expenditures or how exposure to spillover effects influences patent behavior over time.

Table 3

Hausman Test

```
data: patents ~ log_RD + log_spillover
chisq = 44.767, df = 2, p-value = 1.901e-10
alternative hypothesis: one model is inconsistent
```

Note. The Hausman test compares the fixed effects and random effects models to evaluate the consistency of the random effects estimator. The null hypothesis is that the random effects model is consistent and efficient, while the alternative suggests inconsistency due to correlation between regressors and unobserved effects. A highly significant test statistic ($\chi^2 = 44.77$, $df = 2$, $p < .001$) provides strong evidence in favor of the fixed effects model.

To select the appropriate panel data model, I conducted a Hausman test to compare the fixed-effects model with the random-effects model (Shown in Table 3). The test results showed a significant chi-square statistic ($\chi^2 = 44.77$, $p < 0.001$), strongly rejecting the null hypothesis that the explanatory variables are uncorrelated with individual-specific effects. This indicates that the random-effects model is inconsistent in this study, and the fixed-effects model is more appropriate.

Table 4

Hausman Test

```
data: patents ~ log_RD + log_spillover
chisq = 6.9663, df = 2, p-value = 0.03071
alternative hypothesis: one model is inconsistent
```

Note. This Hausman test evaluates whether the instrumental variable (2SLS) model yields significantly different

coefficient estimates compared to the fixed effects model. A statistically significant result ($\chi^2 = 6.97$, $df = 2$, $p = .031$) suggests that the two models are not equivalent, implying potential endogeneity in the explanatory variables that the IV model corrects for.

To further test the necessity of using the instrumental variable (IV) model, I compared the estimation results of the fixed-effects model with the 2SLS instrumental variable model using the Hausman test (See Table 4). The test results showed a chi-square statistic of $\chi^2 = 6.97$, with 2 degrees of freedom and a p-value of 0.031, rejecting the null hypothesis at the 5% significance level. This result reinforces our economic theoretical expectations: both firms' R&D investment and knowledge spillover not only affect patent output but may also be influenced by unobserved factors. Therefore, using the 2SLS instrumental variable model is reasonable and necessary.

In summary, R&D expenditures have a positive contribution to innovation across all specifications, although this effect is overestimated in the naive pooled regression. Technology spillovers are also important drivers of patent production, but this effect is masked in the pooled OLS due to unobserved confounding. The FE model emphasizes the importance of firm-level dynamics, while the IV model ensures that simultaneity bias does not drive the observed relationships. Thus, the core findings are robust across specifications, but their magnitude and importance are sensitive to the modeling approach.

Conclusions, Criticism, and Limitations

We estimated and compared four econometric models: Pooled Ordinary Least Squares (OLS), Fixed Effects (FE), Random Effects (RE), and Two-Stage Least Squares Instrumental Variables (2SLS-IV). In all models, $\log(\text{R\&D})$ shows a strong, positive, and statistically significant impact on patents, confirming the traditional expectation that internal innovation investment leads to increased patent activity. The logarithm of spillovers also shows a significantly positive coefficient in all models except Pooled OLS. This suggests that spillover effects do contribute to firms' patent productivity when unobserved heterogeneity and endogeneity are properly accounted for. The Pooled OLS model provides exaggerated estimates for both R&D and spillover effects, likely due to bias introduced by ignoring firm-specific unobserved heterogeneity. Second, the Hausman test strongly rejects the Random Effects model in favor of Fixed Effects, indicating that unobserved firm-level factors are correlated with regressors and must be controlled for through within transformation. While the 2SLS-IV model considers potential endogeneity of R&D and spillovers using lagged instruments, its coefficient values are slightly smaller but still significant compared to Fixed Effects. Overall, the 2SLS-IV and Fixed Effects models are the most credible specifications, providing complementary insights.

This study still has many limitations. First, both spillover effects and R&D expenditures are measured using logarithms, which may not fully capture the underlying structures. For example, the spatial spillover index may not fully reflect the complexity of knowledge diffusion. The findings are based on a sample of 181 firms from specific industries and countries. Without further testing, generalizing to broader industries or global firms may be invalid. Moreover, the assumption of log-linear relationships may oversimplify potential nonlinear innovation dynamics. Since none of the models include cross-product or higher-order parameters, this may weaken the models' predictive power. Finally, although the study identifies positive spillover and R&D effects, it does not explore industry or firm size heterogeneity, or differences across industries and regions, which may lead the model to miss many important predictors.

For future research, it is recommended to apply more advanced bias-corrected dynamic panel models (such as the carefully instrumented Arellano-Bond GMM model), incorporate interaction terms (e.g., between R&D and spillovers) to capture complementarities. In summary, this study shows the complexity of evaluating innovation production functions and the importance of addressing unobserved heterogeneity and endogeneity when assessing R&D expenditures and spillovers. Moreover, future research on the relationship between corporate innovation output and innovation input should not be limited to model selection and optimization. Instead, more emphasis should be placed on exploring potential factors influencing innovation and indicators reflecting innovation, while striving to collect more diverse data.

Appendix

```
library(dplyr)
library(plm)
library(stargazer)
library(psych)

rm(list = ls())
df <- read.table("D:/AAA/UPennHW/STAT5210/Project/resources/data.mc", header =
FALSE);
colnames(df) <- c("firm_id", "sector", "region",
                  paste0("patents_", 1983:1991),
                  paste0("log_RD_", 1983:1991),
                  paste0("log_spillover_", 1983:1991))

# See Descriptive statistics
desc_stats <- describe(df)
print(desc_stats)
stargazer(df, type = "text", title = "Descriptive Statistics", digits = 2,
          summary.stat = c("mean", "sd", "min", "median", "max"))
```

```

long_df <- reshape(df,
                    varying = list(
                        patents = paste0("patents_", 1983:1991),
                        log_RD = paste0("log_RD_", 1983:1991),
                        log_spillover = paste0("log_spillover_", 1983:1991)
                    ),
                    v.names = c("patents", "log_RD", "log_spillover"),
                    timevar = "year",
                    times = 1983:1991,
                    direction = "long",
                    idvar = "firm_id")

pdata <- pdata.frame(long_df, index = c("firm_id", "year"))

model_within <- plm(patents ~ log_RD + log_spillover,
                    data = pdata,
                    model = "within")

summary(model_within)

#Hausman test (random effects vs fixed effects)
#To compare whether the estimates of random effects models and fixed effects models
are significantly different.
model_random <- plm(patents ~ log_RD + log_spillover, data = pdata, model =
"random")
phtest(model_within, model_random)

summary(model_random)

# 2SLS
pdata$lag_log_RD <- ave(pdata$log_RD, pdata$firm_id, FUN = function(x) c(NA, x[-
length(x)]))
pdata$lag_log_spill <- ave(pdata$log_spillover, pdata$firm_id, FUN = function(x)
c(NA, x[-length(x)]))

pdata_iv <- na.omit(pdata)

# IV regression
model_iv <- plm(patents ~ log_RD + log_spillover | lag_log_RD + lag_log_spill +
log_spillover,
                data = pdata_iv,
                model = "within")

```

```
summary(model_iv)
```

```
#Hausman test (instrumental variables vs fixed effects)
```

```
#To compare whether the estimates of IV models and non-IV models are significantly different.
```

```
phptest(model_within, model_iv)
```

```
# Pooled OLS
```

```
pool_ols <- plm(patents ~ log_RD + log_spillover,  
               data = pdata_iv,  
               index = c("firm_id", "year"),  
               model = "pooling")
```

```
summary(pool_ols)
```

```
# create a table for comparison
```

```
stargazer(pool_ols, model_within, model_random, model_iv,  
          type = "text",  
          title = "Comparison of Panel Data Models",  
          column.labels = c("Pooled OLS", "Fixed Effects", "Random Effects",  
"2SLS-IV"),  
          dep.var.labels = "Patents",  
          keep.stat = c("n", "rsq", "adj.rsq", "f"),  
          no.space = TRUE)
```

```
summary(pool_ols)
```

```
summary(model_random)
```

```
summary(model_within)
```

```
summary(model_iv)
```

Citation

Cincera, M. (1997): PATENTS, R&D, AND TECHNOLOGICAL SPILLOVERS AT THE FIRM LEVEL: SOME EVIDENCE FROM ECONOMETRIC COUNT MODELS FOR PANEL DATA (replication data). Version: 1. Journal of Applied Econometrics. Dataset. <https://doi.org/10.15456/jae.2022313.1256653867>
<https://journaldata.zbw.eu/dataset/patents-rampd-and-technological-spillovers-at-the-firm-level-some-evidence-from-econometric-count-m>

<https://www.jstor.org/stable/2285249?seq=7>